Accident in a Co-generation Plant

Analysis of the Human and Organisational Factors (HOF) involved

Minor Assignment | Technology Risk Governance

MSc in Cyber Risk Strategy and Governance | Prof. Paolo Trucco

Team members

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Working version with all nine events of the time-line (Events 1 and 3 from the class discussion, Events 2, 4–9 own analysis).

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Approach and reading key

Method

Each event of the accident time-line is analysed with the **behavioural approach** (Reason's model of organisational accidents), starting from the *phenotype* (how the event manifests) and going back to the *genotypes* (local conditions and latent factors). For every event we report:

- **one page** with the template compiled in bullet form, following the scheme used in class for Events 1 and 3: human factors pathway on the left, system factors pathway on the right, organisational latent errors on top;
- **one page** of justification, explaining which information of the case text supports each box, the classification of the active error and a short **cognitive view (CREAM)** of the human contribution.

Events 1 and 3 reproduce and complete the solution discussed in class; the other events are our own analysis.

How to read the templates

- **Event**: the event occurs if at least one of the two is present: a failure of a technical defence, barrier or safeguard, or an active human error.
- Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it, with the technical latent error behind it). **White boxes**: contributing conditions.
- **Top box vs. side boxes** (scope of influence): a latent factor that affects only one pathway goes into that pathway's box; a factor that affects both the human and the technical side goes on top.

Sources

[§x.y] section of the case document; [Fig. 3] event time-line; [Class] discussion of the case in class; [Inf.] our inference, not stated explicitly in the case.

Event time-line and overview

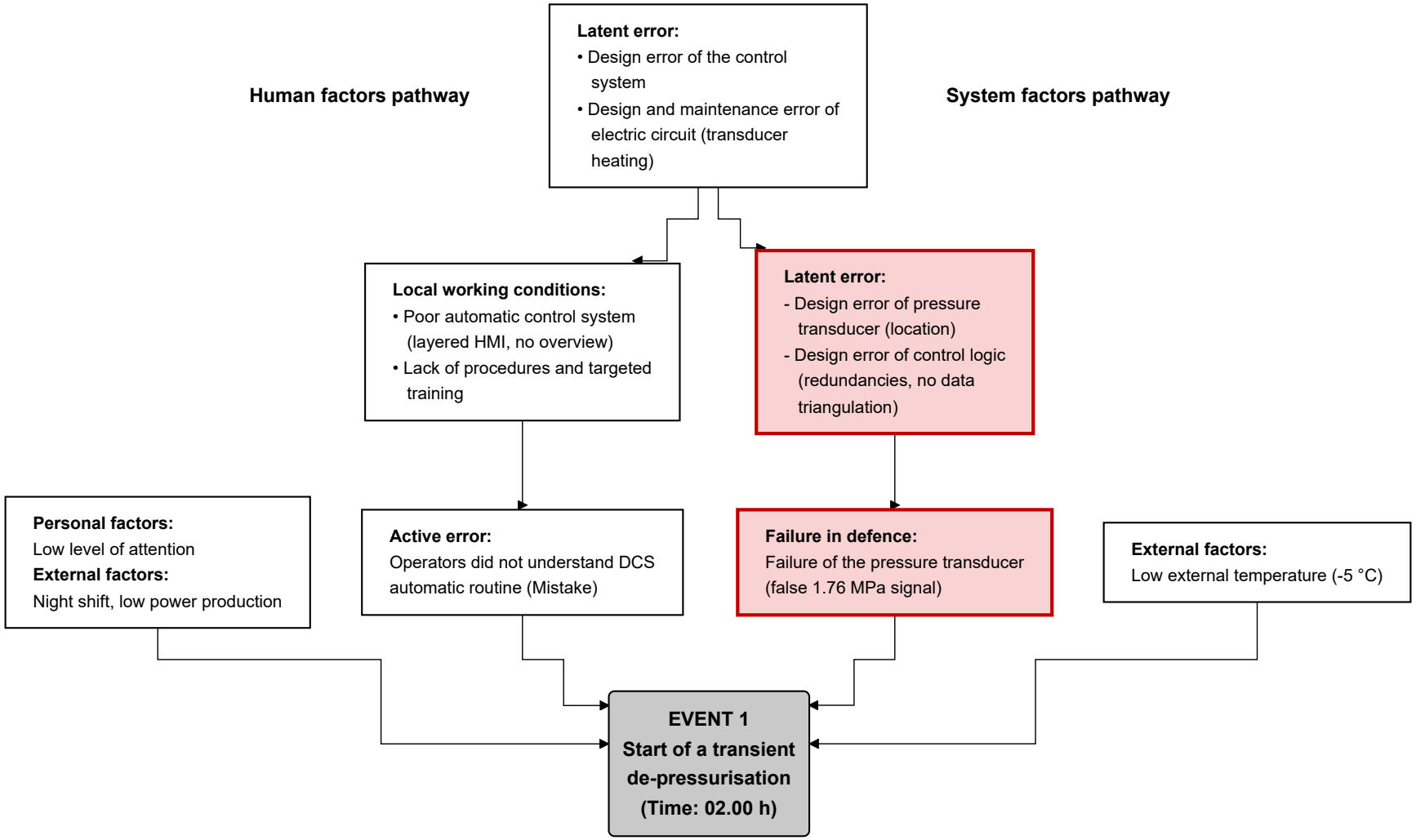
#	Time	Event	Dominant pathway
1	02.00	Pressure transducer failure, start of de-pressurisation	System
2	02.04	Manual activation of main circulation pumps	Human + system
3	02.41	Opening of relief valves, overflow of storage tank, flooding of pumping room	Human + system
4	03.00	Manual closure of relief valves (very late)	Human + system
5	03.20	Fire protection intervention, remote control out of service	System
6	03.45	Temperature increase at heat-exchanger outlet	Human + system
7	04.39	Circulation pumps started by-passing low-pressure protection	Human (violation) + system
8	05.15	Manual start-up of heat recovery from turbo-gas	Human + system
9	06.15	Shock wave (water hammer)	Final event

Emergency phases (causal and contributing factors)

- **From the control room**: 1st de-pressurisation and vaporisation (Events 1, 2, with 4).
- **On the field**, after the loss of remote control: 2nd de-pressurisation (3, 5), vaporisation due to heat exchange (4, 6, 8) and peak pressurisation (7), all converging on the shock wave (9).

Event 1 | Start of a transient de-pressurisation (02.00 h)

The outdoor pressure transducer freezes during a cold winter night and sends a false reading of 1.76 MPa while the real pressure is normal. The automatic control system reacts to the false signal: pressurising pumps slowed down, relief and expansion-tank valves opened, main feed pumps stopped. A rapid de-pressurisation starts.



Active error: Omission (Swain & Guttman) | Mistake (Reason) | Knowledge-based (SRK) | CREAM: Timing → Omission. **Dominant pathway**: system (technical failure of the pressure measurement).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Template reproduces the class solution; items in parentheses complete it with the class discussion.

Event 1 | Justification of the template

Evidence from the case, box by box

Failure in defence (red). The transducer, installed outdoors near an expansion tank, froze together with its signal piping in a cold but "not exceptional" night and sent 1.76 MPa while the real pressure was normal [§2, ev. 1]. **External factors:** low external temperature, about -5 °C [Class].

Latent error, system (red). A safety-critical measurement was placed outdoors, protected only by an electric heating circuit (*location*) [§2; Class]. One single transducer drove both pump speed and pressure valves [§2]: no redundant source and no *data triangulation*, although 1.76 MPa was thermodynamically incompatible with the measured temperature and flow [Class]. The heater failure was also invisible, because the monitoring of the heaters' power supply was wrongly designed [§3.3].

The DCS then did exactly what it was designed for, but on wrong data: it acted as a barrier against a non-existent over-pressure. This is a loss of data *integrity*: the same scenario could be triggered by a cyber attack on the sensor signal [Class].

Active error (white). Operators did not understand what the automatic routine was doing and lost the chance to build situational awareness (*mistake*). Evidence comes from the phenotype of Event 2: their first action was to restart the circulation pumps [Class]. The box is white because the error did not create the de-pressurisation; it removed an early recovery opportunity.

Local working conditions. HMI made of layered windows, without an overview, and no warning for an "abnormal measure" out of the transducer range [§3.3]; no emergency procedures [§3.4]; training only on the job [§3.7].

Personal and external factors. Night shift with base-load demand: almost no interaction between operators and system, hence low attention and low readiness when the transient started [§3.6; Class].

Latent error, organisational (top). Design error of the control system and design & maintenance error of the electric circuit influence *both* pathways: they make the barrier fail and leave operators unable to understand it. Maintenance is also organisational: not providing procedures and training means not maintaining the operators' situational awareness as a defence [Class].

Classification of the active error

Phenotype (Swain & Guttman)	Omission: no diagnosis of, and no intervention on, the automatic routine.
Reason's taxonomy	Mistake: wrong understanding of the situation at the intention stage.
SRK level (Rasmussen)	Knowledge-based (diagnose a DCS acting on false data); the rule-based support was missing (no procedure for this transient).

Cognitive view (CREAM, Hollnagel 1998)

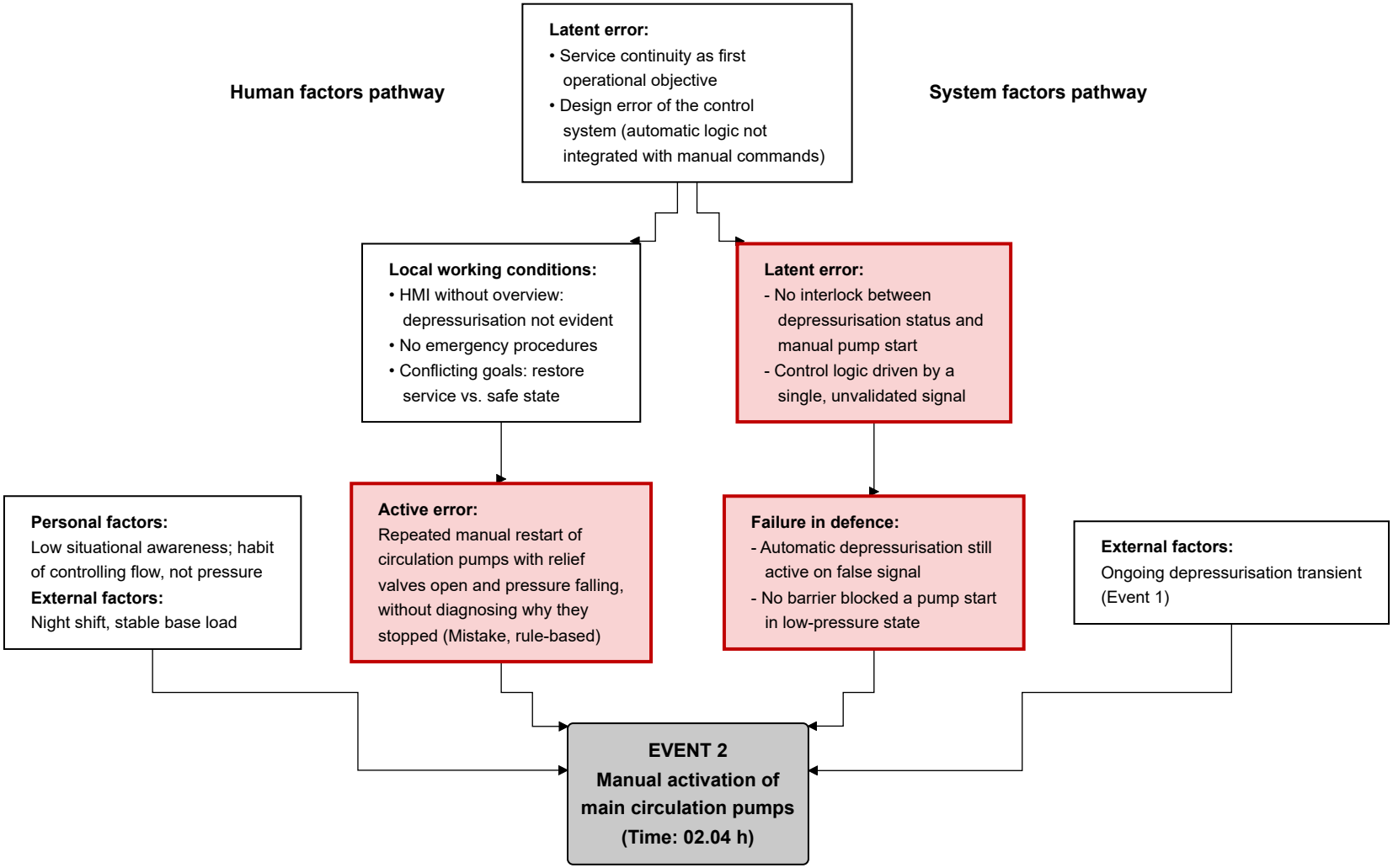
Socio-technical context	Organisation: no emergency procedures, shift team undersized for emergencies. Technology: layered HMI, no out-of-range alarm. Environment: night shift, stable base load.
Cognitive function	Observation → Interpretation (the automatic actions were not noticed or not interpreted).
Individual factors	Psychological aspects (low attention at night); experience and skill (on-the-job training only).
Error mode	Timing → Omission.
General antecedents	Inattention; Observation missed; Faulty diagnosis; Inadequate procedure.
Specific antecedent	None (first event of the sequence).

Improvement levers

- **Prevention**: relocated and redundant pressure sensors with plausibility checks (pressure vs. temperature and flow); alarm on heater supply failure.
- **Recovery**: out-of-range ("abnormal measure") alarm and an overview display of the automatic actions in progress.
- **Emergency management**: procedure and training for the de-pressurisation transient; integrity monitoring of OT sensor data (cyber).
- After the accident the company relocated and duplicated the sensors and replaced most of the DCS [Class].

Event 2 | Manual activation of main circulation pumps (02.04 h)

Although the real pressure is falling, the main recirculation pumps are manually started from the control room at minimum speed. With the relief valves still open, water is pushed out of the system and vaporisation increases.



Active error: Commission, wrong action (Swain & Guttman) | Rule-based mistake (Reason) | Rule-based (SRK) | CREAM: Timing → Too early. **Dominant pathway**: human, enabled by the automatic logic still acting on the false signal.

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis.

Event 2 | Justification of the template

Evidence from the case, box by box

Event and external factors (system). Four minutes after the transducer failure the main recirculation pumps were manually started from the control room, at minimum speed, "even though the pressure in the system was low", with a further trend of vaporisation [§2, ev. 2]. The de-pressurisation transient of Event 1 was still in progress [§2, ev. 1]. In class the restart was described as repeated, with the relief valves open, so that water was pumped out of the system [Class].

Failure in defence (red). The automatic logic kept executing the de-pressurisation on the false 1.76 MPa signal (pressurising pumps slowed, relief and expansion-tank valves open) [§2, ev. 1], and no barrier prevented a manual command that was inconsistent with the plant state [Inf.].

Latent error, system (red). No interlock or permissive linking the status of the relief valves (de-pressurisation in progress) to the start of the pumps [Inf.]; control logic driven by one single, unvalidated signal [§2; Class].

Active error (red). The pumps were restarted without diagnosing why they had stopped. Operators normally follow the heat demand by regulating the flow, not the pressure, so seeing the pumps stopped they applied the familiar rule "pumps stopped → restart them" to a situation where it did not fit [Class]. This is the visible phenotype of the Event 1 mistake.

Local working conditions. Layered HMI without an overview, so the de-pressurisation was not evident [§3.3]; no emergency procedures [§3.4]; conflicting objectives "restore the service" vs. "reach nominal safe conditions" [§3.4, tasks]; limited time [§3.5].

Personal and external factors. Low situational awareness after a long, stable night shift and the habit of controlling flow, not pressure [§3.6; Class].

Latent error, organisational (top). The first, clearly stated objective was to assure heating to the users [§3.1]: it shapes the operators' priorities (human side) and a control system designed around service continuity, where automatic protections and manual commands are not integrated (system side). Hence it influences both pathways.

Classification of the active error

Phenotype (Swain & Guttman)	Commission: an action was performed, but it was the wrong one for the real plant state.
Reason's taxonomy	Mistake, rule-based: correct execution of an inappropriate rule (wrong intention).
SRK level (Rasmussen)	Rule-based: a "strong but wrong" rule replaced the diagnosis that the novel situation required.

Cognitive view (CREAM, Hollnagel 1998)

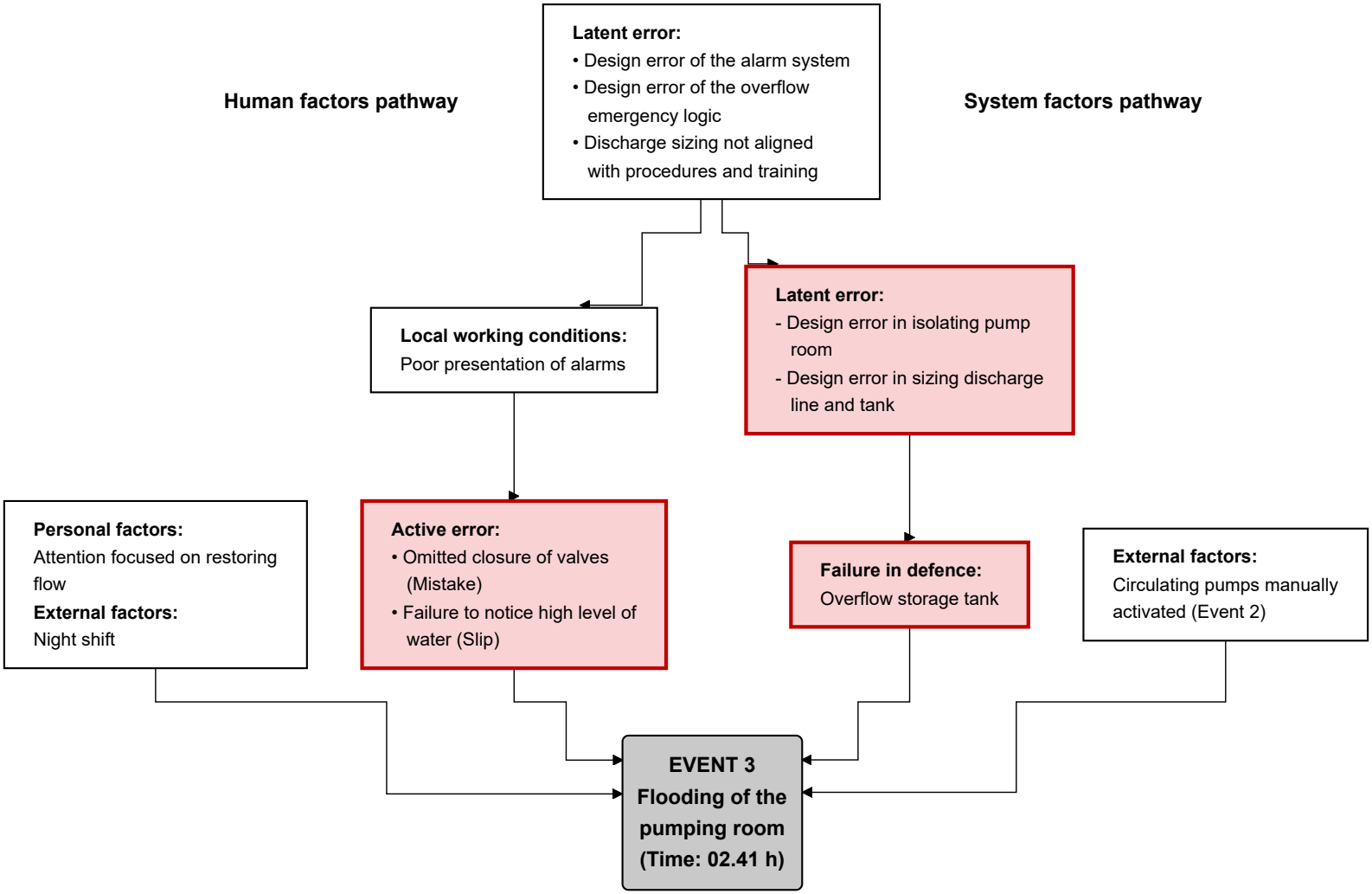
Socio-technical context	Organisation: service continuity as first objective, no emergency procedures. Technology: no overview of automatic actions, false pressure reading. Environment: night shift, base load.
Cognitive function	Interpretation → Planning (wrong goal: restore flow instead of reaching a safe state).
Individual factors	Experience and skill (habit of flow control); psychological aspects (low situational awareness).
Error mode	Timing → Too early (pumps restarted before the pressure was restored and the valves closed).
General antecedents	Faulty diagnosis; Inadequate procedure.
Specific antecedent	Earlier omission (no diagnosis at Event 1).

Improvement levers

- **Prevention**: permissive on pump start when relief valves are open or pressure is below a threshold.
- **Recovery**: procedure step "identify why the pumps tripped before any restart"; display of the active automatic sequence.
- **Emergency management**: explicit goal hierarchy (safe state before service) in procedures and training.

Event 3 | Flooding of the pumping room (02.41 h)

The storage tank reaches its maximum level and water is discharged through the relief valves. The emergency discharge system is not dimensioned for this flow and boiling water starts to flood the pumping room.



Active errors: Omission (Swain & Guttman) | Mistake + Slip (Reason) | Skill-based (SRK) | CREAM: Timing → Omission. **Dominant pathway**: human and system together (alignment between technical and organisational design).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Template reproduces the class solution; the personal/external factors box and the third top item complete it with the class discussion.

Event 3 | Justification of the template

Evidence from the case, box by box

Event and external factors (system). The maximum level in the storage tank resulted in water discharge through the relief valves; the emergency system was not dimensioned for the flow and boiling water flooded the pumping room [§2, ev. 3]. The circulation pumps manually activated in Event 2 pushed out a flow 4–5 times higher than any value considered by the designers [Class].

Failure in defence (red). Overflow of the storage tank: the last barrier collecting the discharged water was exceeded [§2, ev. 3].

Latent error, system (red). Discharge line and tank were sized for the normal steam condensate released by safety valves, not for the full flow of the pumps; the pumping room was not isolated from the discharge [Class].

Active errors (red). (1) *Omitted closure of the valves* (mistake): with relief valves open, checking whether the pressure is really high and closing the valves before restarting the pumps is a basic operation. (2) *Failure to notice the high water level* (slip): the alarm was available but not perceived [Class].

Local working conditions. The "high level of water in the storage tank" alarm was shown, but dispersed in a long list of less important alarms; HMI made of layered windows with no general overview [§3.3]; no video surveillance of the rooms [§3.3].

Personal and external factors (completion). Night shift [§3.6]; attention focused on restoring the flow, as shown by Event 2 [Class].

Why no "lack of training" here. Both errors are at the skill-based level: basic competences acquired with entry-level training and experience, which must be supported by a clear alarm presentation. Training is decisive for Event 1 (rule- and knowledge-based); it may be added here only if specified as entry-level training [Class].

Latent error, organisational (top). Design errors of the alarm system and of the overflow emergency logic. The size of the tank and the operators' procedures and training were decided separately: a tank is large enough only if operators can detect and close the valves *before* it fills. This misalignment between technical and organisational design affects both pathways, so it goes on top [Class].

Classification of the active error

Phenotype (Swain & Guttman)	Omission: required action not performed (valve closure); omission of detection (alarm not perceived).
Reason's taxonomy	Mistake (omitted closure, wrong intention) + Slip (alarm missed at execution/perception level).
SRK level (Rasmussen)	Skill-based: elementary routine operations, like changing gear while driving.

Cognitive view (CREAM, Hollnagel 1998)

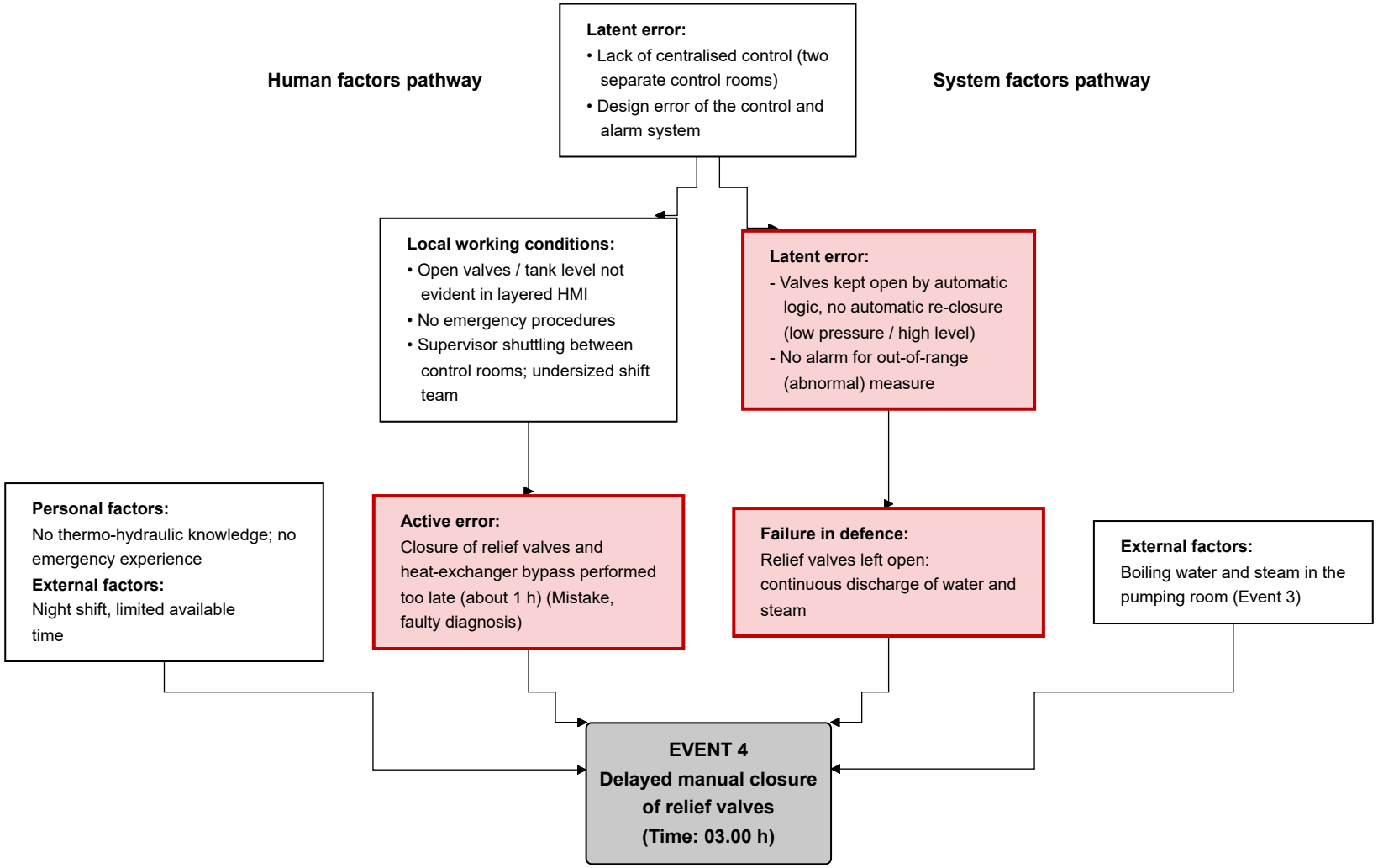
Socio-technical context	Organisation: no procedures, team undersized for emergencies, tank sizing not aligned with operator response time. Technology: alarm flooding, layered HMI, no CCTV. Environment: night shift.
Cognitive function	Observation (high-level alarm missed) → Planning (closure not planned).
Individual factors	Psychological aspects (attention narrowed on flow restoration).
Error mode	Timing → Omission.
General antecedents	Observation missed; Inattention.
Specific antecedent	Earlier omission (no diagnosis of the de-pressurisation in Events 1–2).

Improvement levers

- **Prevention**: size discharge line and tank jointly with procedures, comparing time-to-overflow with time-to-detect-and-act; isolate the pumping room.
- **Recovery**: alarm prioritisation (no long undifferentiated lists), overview display, CCTV of critical rooms.
- **Emergency management**: entry-level skills checked periodically, not only on the job.

Event 4 | Delayed manual closure of relief valves (03.00 h)

The manual closure of the relief valves and the by-pass of the heat exchanger, essential protection actions, are performed with a very substantial delay: about one hour after the valves opened (02.00 h) and almost twenty minutes after the overflow started (02.41 h).



Active error: Commission, too late (Swain & Guttman) | Mistake, faulty diagnosis (Reason) | Knowledge-based (SRK) | CREAM: Timing → Too late. **Dominant pathway**: human and system together.

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis.

Event 4 | Justification of the template

Evidence from the case, box by box

Event. “Manual closure of relief valves and bypass of the heat exchanger, i.e. essential protection actions, were performed with a very substantial delay” [§2, ev. 4]. The time-line places the closure at 03.00, after the opening at 02.00 and the overflow at 02.41 [Fig. 3].

Failure in defence (red). The relief valves, a protection against over-pressure, stayed open in a situation of *under*-pressure and kept discharging water and steam [§2, ev. 1 and 3].

Latent error, system (red). The automatic logic held the valves open on the false signal and had no automatic re-closure on low pressure or high tank level [Inf.]; no warning for an out-of-range (“abnormal”) measure [§3.3].

External factors (system). Boiling water and steam in the pumping room, where part of the actions had to be carried out [§2, ev. 3; Inf.].

Active error (red). The closure was performed, but too late: operators did not recognise in time the real state (open valves, falling pressure, filling tank). It is a mistake rooted in a faulty diagnosis, not a forgotten step.

Local working conditions. Open valves and tank level not evident in the layered HMI, high-level alarm lost in a long list [§3.3]; no emergency procedures [§3.4]; the supervisor had to move between two control rooms not reachable in a few minutes [§3.2]; shift team insufficient for an emergency [§3.1].

Personal and external factors. Operators had no thermo-hydraulic knowledge [§3.1] and inadequate experience for emergencies [§3.7]; limited time for recovery actions [§3.5]; night shift [§3.6].

Latent error, organisational (top). Lack of centralised control, with misunderstandings and incoherent decisions between the two control rooms [§3.1]: on the human side it splits information and decisions, on the technical side the two units have no integrated view or logic. The design error of the control and alarm system also acts on both sides (wrong valve logic and poor information to operators).

Classification of the active error

Phenotype (Swain & Guttman)	Commission: the action was performed, but too late (timing error).
Reason’s taxonomy	Mistake: wrong situation assessment (faulty diagnosis) delayed the correct intention.
SRK level (Rasmussen)	Knowledge-based: an unfamiliar transient had to be diagnosed without procedures.

Cognitive view (CREAM, Hollnagel 1998)

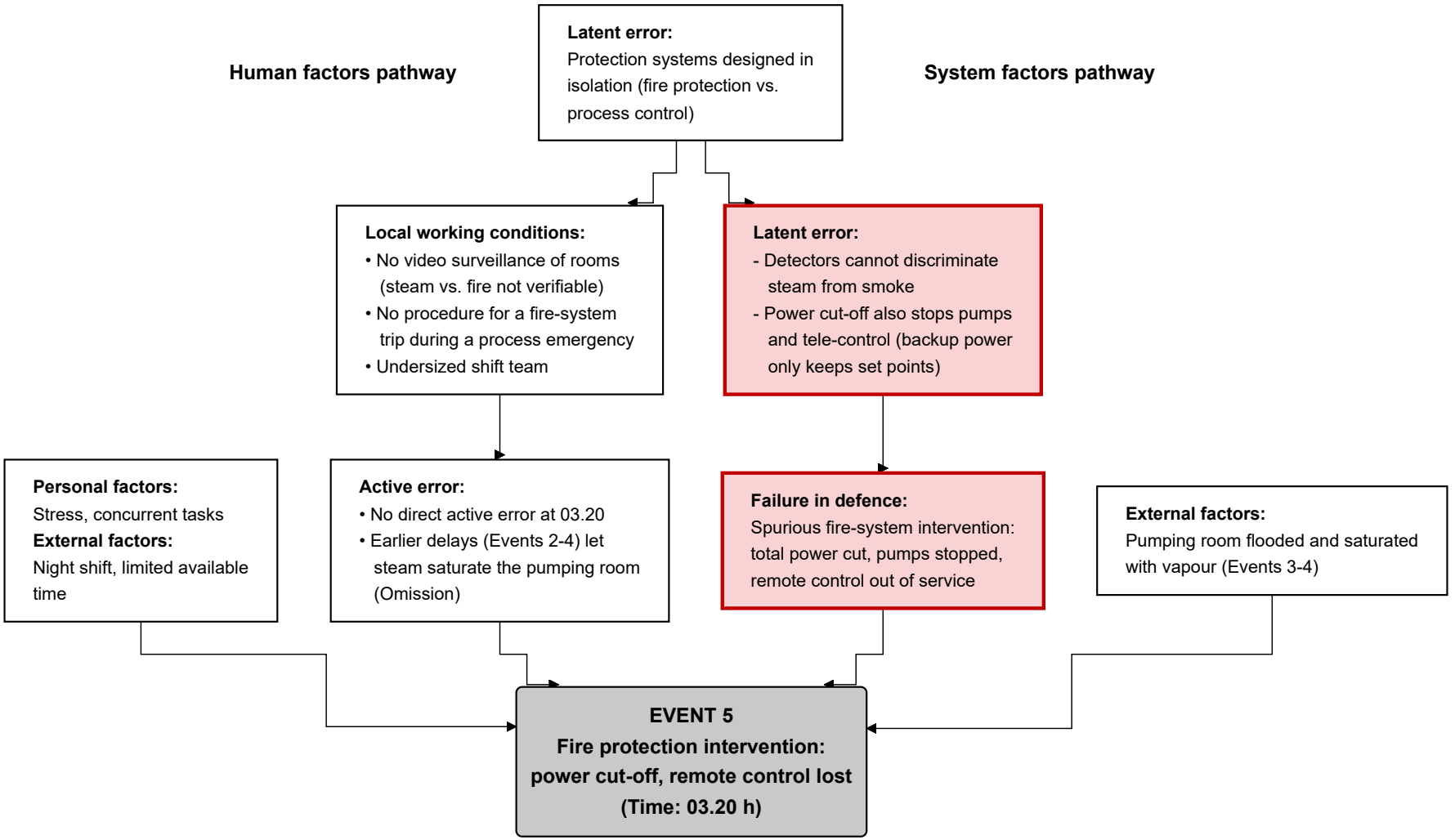
Socio-technical context	Organisation: two separate control rooms, supervisor shuttling, undersized team, no procedures. Technology: layered HMI, alarm flooding, no out-of-range alarm. Environment: night shift, flooded pumping room.
Cognitive function	Interpretation (plant state misjudged).
Individual factors	Experience and skill (no thermo-hydraulic knowledge, no emergency practice).
Error mode	Timing → Too late.
General antecedents	Faulty diagnosis; Observation missed; Inadequate procedure.
Specific antecedent	Earlier omission (state not diagnosed in Events 1–3).

Improvement levers

- **Prevention**: automatic re-closure of relief valves on low pressure or high tank level.
- **Recovery**: alarm on “relief valve open for more than X minutes”; overview display of valve states.
- **Emergency management**: emergency procedure with time targets; single control room for generation and network (adopted after the accident [Class]).

Event 5 | Fire protection intervention, remote control out of service (03.20 h)

Flooding of the pumping room and saturation of the atmosphere with vapour trigger the fire protection system, which cannot distinguish vapour from smoke. All electrical power is cut off, the pumps stop and the remote control system goes out of service.



Active error: no direct active error at 03.20 (indirect omission / too late from Events 2–4) | CREAM: Timing → Too late. **Dominant pathway**: system (spurious intervention of a safety system), as in Event 1.

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis.

Event 5 | Justification of the template

Evidence from the case, box by box

Event and external factors (system). "Flooding of the pump room and saturation of atmosphere with vapour caused the intervention of the fire protection system, which was unable to distinguish between vapour and smoke": complete power cut-off, pumps stopped, remote control out of service [§2, ev. 5]. From here on the emergency had to be managed on the field, with operators "blind" and without remote commands [Class].

Failure in defence (red). Spurious intervention of a safety barrier: a system designed against fire became an initiator of loss of control, removing pumps and tele-control at the same time.

Latent error, system (red). Detectors not able to discriminate steam from smoke [§2, ev. 5]; the fire logic cuts *all* electrical power, and the back-up only keeps the automatic control settings [§2], not the tele-control; pumping room not isolated [Class, ev. 3].

Active error (white). No human action is synchronous with the event. The delays of Events 2–4 let the steam release persist until the detectors tripped: an indirect contribution (omission / too late), so the box is white, as in Event 1.

Local working conditions. No video surveillance to check the rooms, so fire vs. steam could not be verified [§3.3]; no procedure for a fire-system trip during a process emergency [§3.4]; team insufficient for an emergency [§3.1].

Personal and external factors. Concurrent and conflicting tasks [§3.4, tasks]; limited available time [§3.5]; night shift [§3.6]; resulting stress [Inf.].

Latent error, organisational (top). Protection systems designed in isolation: the fire protection logic was not integrated with process safety. On the technical side the power cut disables process barriers; on the human side operators lose information and commands with no procedure to handle it. It is the same alignment problem between different safety (and security) design processes discussed in the course [Inf.].

Classification of the active error

Phenotype (Swain & Guttman)	Not applicable at 03.20. Indirect contribution: omission / too late in containing the steam release (Events 2–4).
Reason's taxonomy	No new unsafe act; consequences of the earlier mistakes.
SRK level (Rasmussen)	Not applicable (no task performed at 03.20).

Cognitive view (CREAM, Hollnagel 1998)

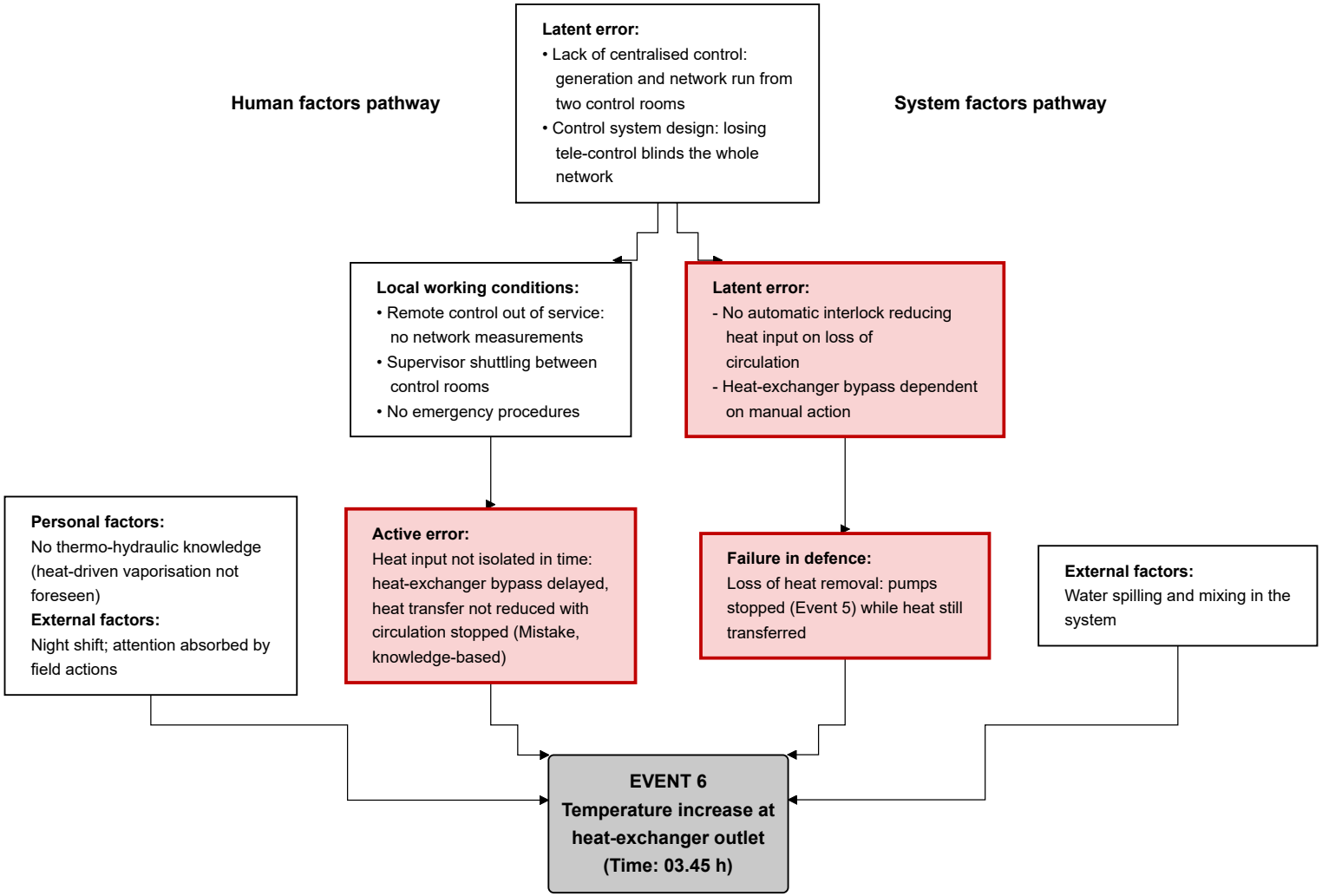
Socio-technical context	Organisation: no procedure for fire-system trips, undersized team. Technology: fire logic cutting all power, back-up keeping only set points, no CCTV. Environment: night shift, vapour-saturated pumping room.
Cognitive function	Observation (state of the pumping room not observable).
Individual factors	Psychological aspects (stress, competing tasks).
Error mode	Timing → Too late (containment of the steam release).
General antecedents	Observation missed; Inadequate procedure.
Specific antecedent	Earlier omission (valves not closed in time, Events 3–4).

Improvement levers

- **Prevention**: multi-criteria fire detection (smoke plus temperature/flame) in rooms exposed to steam; isolation of the pumping room.
- **Recovery**: selective load shedding and UPS preserving tele-control and essential pumps; CCTV of critical rooms.
- **Emergency management**: procedure for loss of remote control (local control via Series 9000 panels) and joint design reviews of fire and process protections.

Event 6 | Temperature increase at heat-exchanger outlet (03.45 h)

The water temperature at the heat-exchanger outlet increases further, as a consequence of water spilling and mixing in the system, while circulation is stopped and heat is still transferred to the loop. Vaporisation due to heat exchange begins.



Active error: Omission / too late (Swain & Guttman) | Knowledge-based mistake (Reason) | Knowledge-based (SRK) | CREAM: Timing → Too late. **Dominant pathway**: system (loss of heat removal), with a human contribution.

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis.

Event 6 | Justification of the template

Evidence from the case, box by box

Event and external factors (system). "Further increase of the water temperature occurred as a consequence of the water spilling and mixing in the system" [§2, ev. 6]; the time-line reports an increase of the temperature at the heat-exchanger outlet at 03.45 [Fig. 3]. In the causal diagram it belongs to the "vaporisation due to heat exchange" phase (Events 4, 6, 8, triggered by 5).

Failure in defence (red). Loss of heat removal: the pumps stopped by the power cut (Event 5) while heat was still transferred to the loop, so the temperature rose and the margin against vaporisation was lost [§2, ev. 5–6; Class].

Latent error, system (red). No automatic interlock reducing the heat input or by-passing the heat exchangers on loss of circulation [Inf.]; the heat-exchanger by-pass depended on a manual action, which came very late [§2, ev. 4].

Active error (red). The heat input was not isolated in time: by-pass of the heat exchanger delayed and heat transfer not reduced while circulation was stopped. Operators had no thermo-hydraulic knowledge to foresee vaporisation due to heat exchange [§3.1]: a knowledge-based mistake.

Local working conditions. Remote control out of service since 03.20, so no network measurements [§2, ev. 5]; the supervisor had to move between the two control rooms [§3.2]; no emergency procedures [§3.4].

Personal and external factors. No thermo-hydraulic knowledge [§3.1]; attention absorbed by field actions in the flooded pumping room [Class; Inf.]; night shift [§3.6].

Latent error, organisational (top). Lack of centralised control: power generation and network were run from control rooms in different buildings [§1.3; §3.1]. It affects the human side (no shared decisions on heat input) and the technical side (no cross-unit protection). The control system design, where losing tele-control blinds the whole network, also acts on both sides.

Classification of the active error

Phenotype (Swain & Guttman)	Omission / too late: heat-exchanger by-pass and reduction of heat input not performed in time.
Reason's taxonomy	Mistake, knowledge-based: the physical consequence of heating a stagnant loop was not foreseen.
SRK level (Rasmussen)	Knowledge-based: novel situation requiring thermo-hydraulic understanding.

Cognitive view (CREAM, Hollnagel 1998)

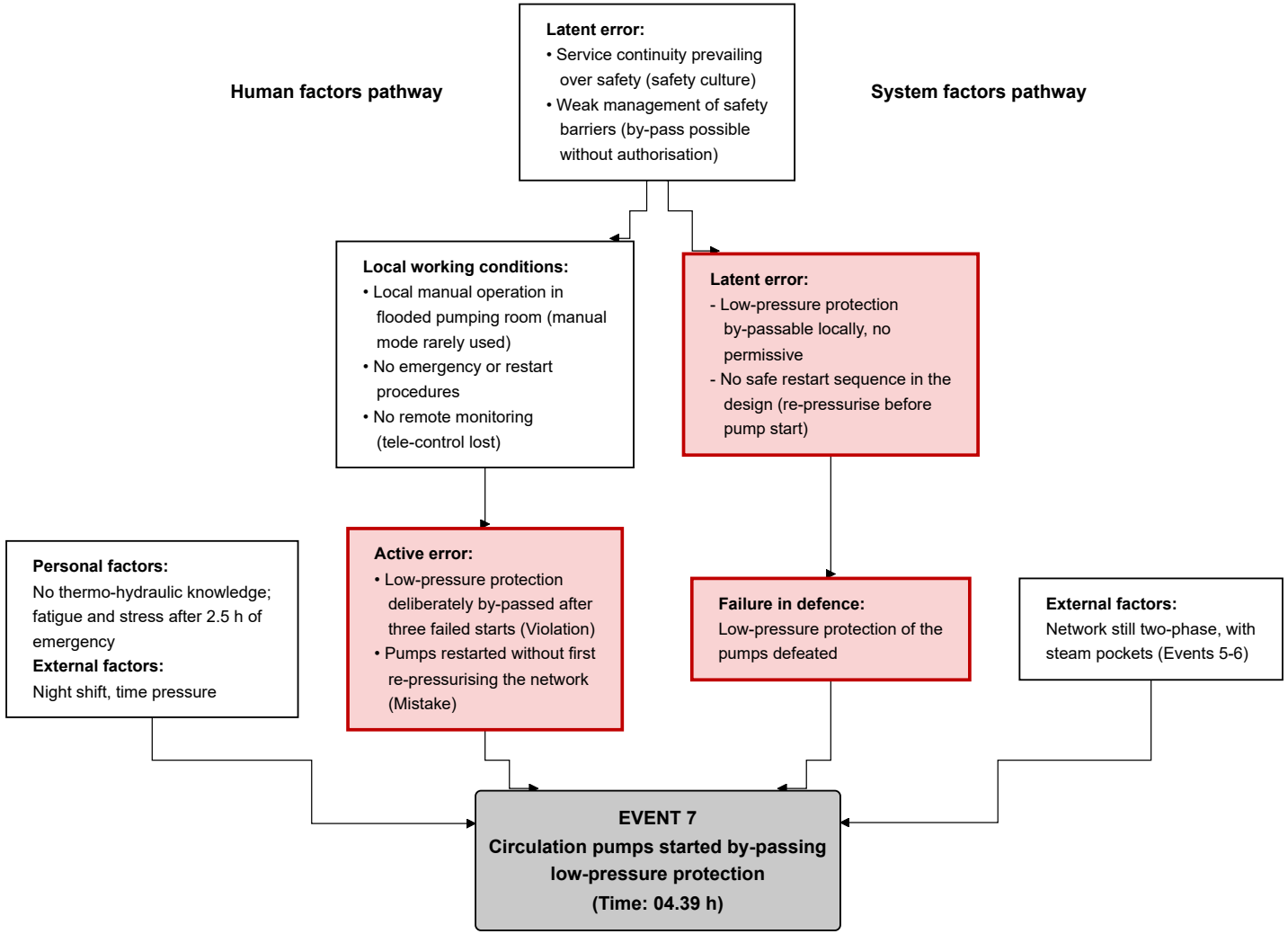
Socio-technical context	Organisation: two control rooms, supervisor shuttling, no procedures. Technology: tele-control lost, manual by-pass, no automatic run-back of heat input. Environment: night shift, flooded pumping room.
Cognitive function	Interpretation → Planning (temperature rise not anticipated, no plan to isolate heat input).
Individual factors	Experience and skill (no thermo-hydraulic knowledge).
Error mode	Timing → Too late.
General antecedents	Communication failure (between control rooms); Faulty diagnosis; Observation missed.
Specific antecedent	Earlier omission (by-pass not performed in Event 4).

Improvement levers

- **Prevention**: automatic run-back of heat input / heat-exchanger by-pass on low network flow.
- **Recovery**: back-up power for monitoring; shared display of network temperatures in both control rooms.
- **Emergency management**: training on two-phase phenomena and single control room (adopted after the accident [Class]).

Event 7 | Circulation pumps started by-passing low-pressure protection (04.39 h)

After three unsuccessful attempts, the recirculation pumps are manually activated by-passing their automatic low-pressure protection. Pressure first drops further and is then followed by a violent pressurisation of the system.



Active errors: Violation (Swain & Guttman) | Violation + knowledge-based mistake (Reason) | Knowledge-based, handled with rules (SRK) | CREAM: Sequence → Jump forward. **Dominant pathway**: human (deliberate defeat of a working barrier).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis.

Event 7 | Justification of the template

Evidence from the case, box by box

Event. “Manual activation of the recirculation pumps, bypassing the low-pressure protection of the pumps (three attempts were made before manual activation) caused a further reduction of pressure, which was followed by a violent pressurisation of the system” [§2, ev. 7]. In the causal diagram this is the “peak pressurisation” branch [Class].

Failure in defence (red). The low-pressure protection of the pumps worked: it blocked three start attempts. It failed as a barrier only because it was defeated by the by-pass.

Latent error, system (red). The protection could be by-passed locally, with no permissive or authorisation [Inf.]; the design offered no safe restart sequence (re-pressurise and vent the network before starting the pumps) [Inf.].

External factors (system). The network was still two-phase, with steam pockets left by the de-pressurisation and by heat exchange (Events 5–6): the precondition for a water hammer on re-pressurisation [§2; Class].

Active errors (red). (1) The protection was *deliberately* by-passed: a violation, i.e. a deliberate but not reprehensible deviation from safe practice, motivated by the will to restore the service. (2) The pumps were restarted without first re-pressurising the network: a knowledge-based mistake, since the risk of collapsing steam pockets was not understood [§3.1].

Local working conditions. Manual operation is an emergency back-up, “rarely utilised for obvious reasons of complexity and difficulty” [§1.2.1], here performed locally in a flooded pumping room [§1.3; §2]; no remote monitoring after Event 5 [§2, ev. 5]; no emergency or restart procedures [§3.4]; conflicting objectives restore service vs. safe state [§3.4, tasks].

Personal and external factors. No thermo-hydraulic knowledge [§3.1]; skills inadequate for emergencies [§3.7]; about 2.5 hours of emergency at night, hence fatigue and stress [Fig. 3; Inf.]; limited time [§3.5].

Latent error, organisational (top). Service continuity as first objective [§3.1] and weak management of safety barriers (by-pass allowed without authorisation): they make the by-pass technically possible and culturally acceptable, so they influence both pathways.

Classification of the active error

Phenotype (Swain & Guttman)	Violation (deliberate deviation from a protection), performed as a commission.
Reason’s taxonomy	Violation (by-pass) + Mistake, knowledge-based (restart without re-pressurisation).
SRK level (Rasmussen)	Knowledge-based situation handled with the rule “restore the flow”.

Cognitive view (CREAM, Hollnagel 1998)

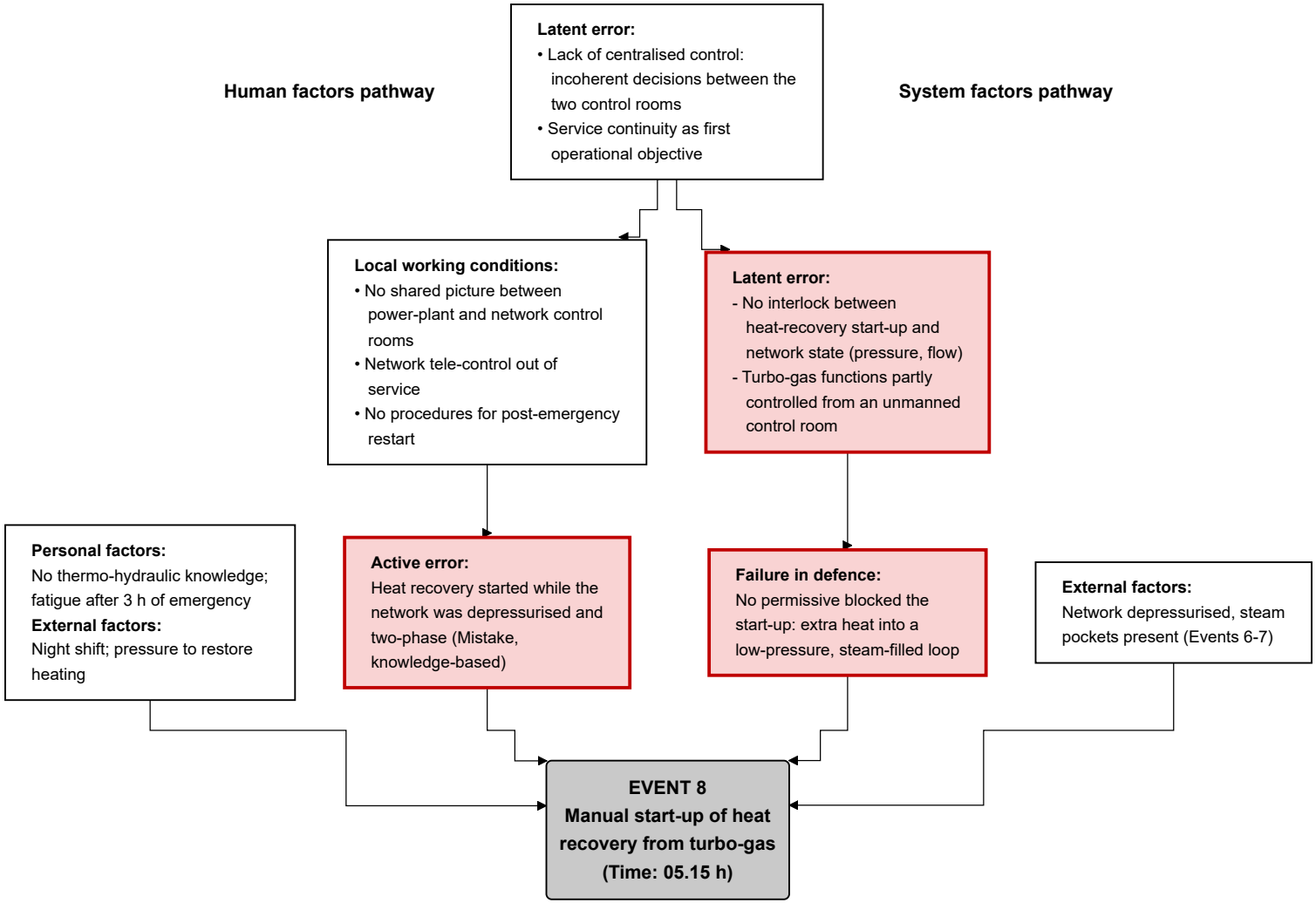
Socio-technical context	Organisation: service priority, no rule on by-passing protections, no procedures. Technology: protection by-passable locally, no remote monitoring. Environment: night, flooded pumping room, manual local operation.
Cognitive function	Planning (restart planned without the re-pressurisation step).
Individual factors	Physiological aspects (fatigue after hours of emergency); experience and skill (no thermo-hydraulic knowledge).
Error mode	Sequence → Jump forward (re-pressurisation and venting skipped).
General antecedents	Faulty diagnosis; Inadequate plan; Inadequate procedure.
Specific antecedent	None identified.

Improvement levers

- **Prevention**: key-locked by-pass requiring supervisor authorisation and logging; hard permissive on network pressure.
- **Recovery**: restart procedure (re-pressurise, vent, slow start) and training on water hammer.
- **Emergency management**: explicit rule “safe state before service” endorsed by management.

Event 8 | Manual start-up of heat recovery from turbo-gas (05.15 h)

The heat recovery system of the turbo-gas group is manually started. Heat is injected into a loop that is still at low pressure and full of steam pockets, feeding a second vaporisation.



Active error: Commission, wrong timing (Swain & Guttman) | Knowledge-based mistake + communication failure (Reason) | Knowledge-based (SRK) | CREAM: Timing → Too early. **Dominant pathway**: human and organisational (coordination between the two control rooms).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis.

Event 8 | Justification of the template

Evidence from the case, box by box

Event. "The heat recovery system from the turbo-gas system was started" [§2, ev. 8]; manual start-up at 05.15 [Fig. 3]. In class: the power-plant operators connected the heat exchanger (condenser), adding heat to a pipe already at low pressure and full of steam [Class].

External factors (system). Network still de-pressurised, with steam pockets after Events 6–7 [§2].

Failure in defence (red). No permissive blocked the start-up: extra heat entered a two-phase loop [Inf.].

Latent error, system (red). No interlock between heat-recovery start-up and network state (pressure, flow) [Inf.]; some functions of the turbo-gas group were controlled from a control room "not normally manned" [§1.3].

Active error (red). The start-up was performed at the wrong moment, with the network not yet in a safe state: a knowledge-based mistake (consequences not foreseen, [§3.1]) combined with a communication failure between the power-plant and the network control rooms.

Local working conditions. Power plant and network controlled from different control rooms in different buildings [§1.3]; lack of centralised control leading to "misunderstanding and incoherent decisions" [§3.1]; the supervisor moving between the two rooms [§3.2]; network tele-control lost [§2, ev. 5]; no procedures for restart after an emergency [§3.4].

Personal and external factors. No thermo-hydraulic knowledge [§3.1]; more than three hours of emergency at night [Fig. 3; Inf.]; pressure to restore heating, the first stated objective [§3.1], possibly reinforced by the approaching morning demand [Inf.].

Latent error, organisational (top). Lack of centralised control between generation and distribution [§3.1]: it causes the coordination failure (human side) and the absence of technical interlocks across the two units (system side). Service continuity as first objective pushes both towards early restart.

Classification of the active error

Phenotype (Swain & Guttman)	Commission: correct action in itself, performed at the wrong time.
Reason's taxonomy	Mistake, knowledge-based, with communication failure between teams.
SRK level (Rasmussen)	Knowledge-based: interaction between two subsystems in an abnormal state.

Cognitive view (CREAM, Hollnagel 1998)

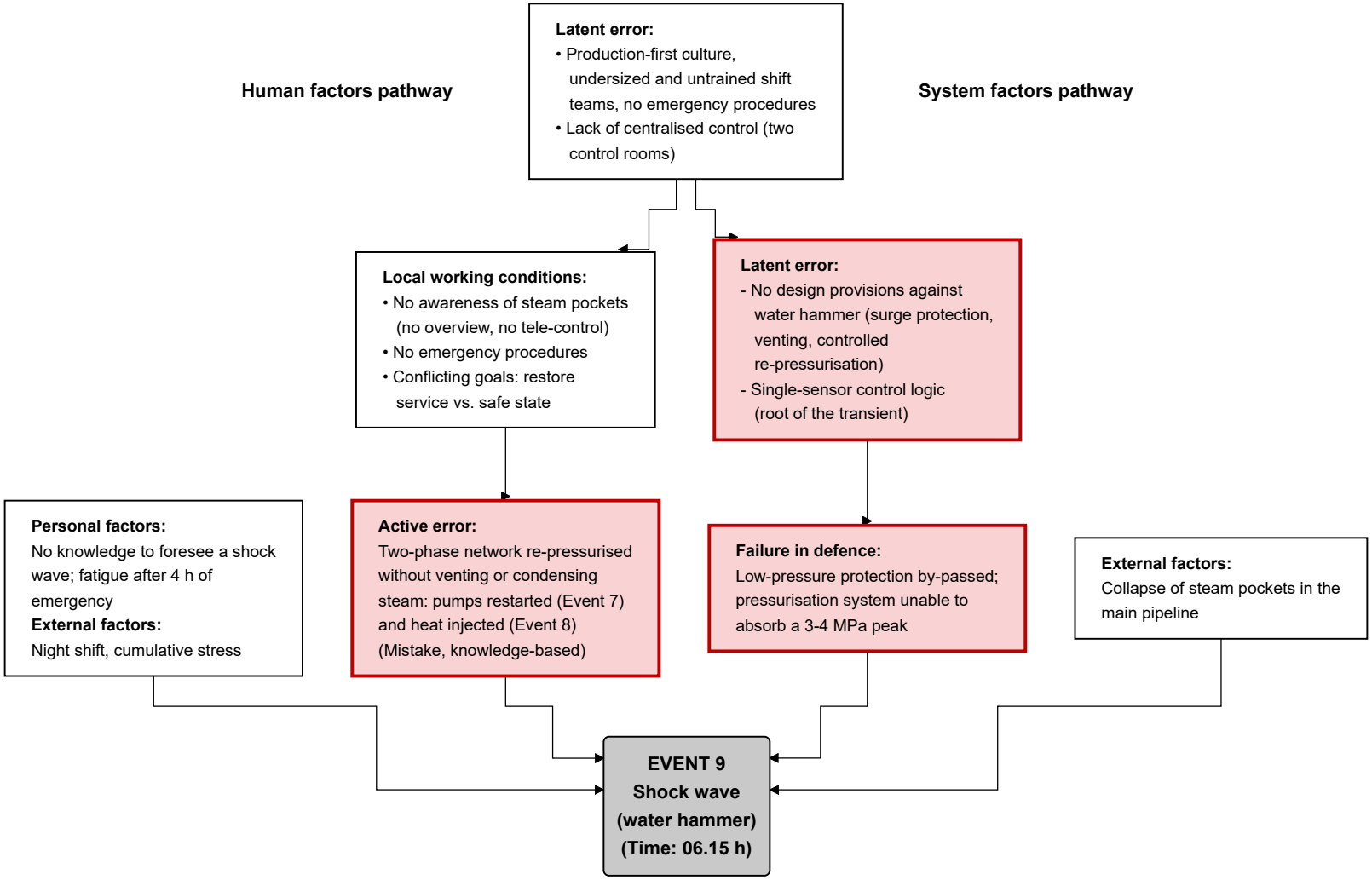
Socio-technical context	Organisation: two control rooms, no centralised control, service priority, no restart procedures. Technology: no cross-unit interlock, network tele-control lost. Environment: night, long emergency.
Cognitive function	Interpretation → Planning (network state unknown or misjudged in the power-plant control room).
Individual factors	Experience and skill (no thermo-hydraulic knowledge); physiological aspects (fatigue).
Error mode	Timing → Too early.
General antecedents	Communication failure; Faulty diagnosis; Inadequate procedure.
Specific antecedent	Earlier omission (network not brought back to a safe state, Events 6–7).

Improvement levers

- **Prevention**: permissive on heat-recovery start-up based on network pressure and flow.
- **Recovery**: shared status display and formal hand-shake between the two control rooms before restarting heat sources.
- **Emergency management**: single control room for generation and network, adopted after the accident [Class].

Event 9 | Shock wave, water hammer (06.15 h)

A very violent shock wave displaces one anchor point of the main pipeline by 90 mm, leaving it permanently deformed (force of about 35–45 t, pressure peak of 3–4 MPa), and reaches the heat exchangers, seriously damaging their internal components.



Active error: Commission (Swain & Guttman) | Knowledge-based mistake, plus the violation of Event 7 (Reason) | Knowledge-based (SRK) | CREAM: Timing → Too early. **Dominant pathway**: both: final alignment of all the holes (organisational accident).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Own analysis. Event 9 summarises the whole accident and largely overlaps with Events 1–8.

Event 9 | Justification of the template

Evidence from the case, box by box

Event. Shock wave displacing an anchor point by 90 mm, force of 35–45 t, instantaneous pressure peak of 3–4 MPa, serious damage inside the heat exchangers [§2, ev. 9]. The condenser had to be replaced five years before the end of its life (about 30% of its useful life lost, on an asset of about 15 MEUR) [Class].

External factors (system). Steam pockets trapped in a two-phase flow collapse when the loop is suddenly pressurised: the liquid fills the voids and generates a local shock wave travelling along the pipe [Class].

Failure in defence (red). All the layers of defence-in-depth were defeated: process margins lost (two-phase fluid), safety systems defeated (low-pressure protection by-passed in Event 7), pressurisation and expansion system unable to absorb a 3–4 MPa peak [§2].

Latent error, system (red). No design provisions against water hammer (surge protection, venting, controlled re-pressurisation) [Inf.]; single-sensor control logic at the origin of the whole transient [§2; Class].

Active error (red). The two-phase network was re-pressurised without venting or condensing the steam: pumps restarted (Event 7) and heat injected (Event 8). The case states that operators lacked the thermo-hydraulic knowledge needed to foresee and prevent a violent shock wave [§3.1]: knowledge-based mistake.

Local working conditions. No awareness of the steam pockets (no overview, no telecontrol) [§3.3; §2, ev. 5]; no emergency procedures [§3.4]; conflicting goals [§3.4, tasks].

Personal and external factors. About four hours of emergency during the night shift [Fig. 3; Inf.]; cumulative stress [Inf.].

Latent error, organisational (top). Production-first culture, undersized and untrained shift teams, no emergency procedures, two separate control rooms [§3.1; §3.4; §3.7]. These factors are not independent: they open aligned holes in technical and human barriers, the Swiss cheese trajectory of an organisational accident.

Classification of the active error

Phenotype (Swain & Guttman)	Commission: re-pressurisation actions performed at the wrong time.
Reason's taxonomy	Mistake, knowledge-based (plus the violation of Event 7).
SRK level (Rasmussen)	Knowledge-based: no rule or skill covered re-pressurising a two-phase network.

Cognitive view (CREAM, Hollnagel 1998)

Socio-technical context	Organisation: production priority, staffing, training and procedures inadequate, two control rooms. Technology: no anti-water-hammer design, no overview. Environment: night, four hours of emergency.
Cognitive function	Interpretation → Planning (presence and danger of steam pockets not understood).
Individual factors	Experience and skill (no thermo-hydraulic knowledge); physiological aspects (fatigue).
Error mode	Timing → Too early (re-pressurisation before steam was removed).
General antecedents	Faulty diagnosis; Communication failure; Inadequate plan.
Specific antecedent	Earlier omission (Events 2–8).

Improvement levers

- **Prevention**: water-hammer protection (surge vessels, venting points, slow-acting valves).
- **Recovery**: controlled re-pressurisation procedure with sub-cooling checks.
- **Emergency management**: integrated emergency organisation, 5-year programme on procedures and retraining (adopted after the accident [Class]).

Cross-event synthesis: recurring latent factors

Latent factor	1	2	3	4	5	6	7	8	9
Single, unvalidated pressure signal driving the control logic	●	●		●					●
HMI without overview, alarm flooding, no out-of-range alarm	●	●	●	●					●
No emergency procedures	●	●	●	●	●	●	●	●	●
On-the-job training only, no thermo-hydraulic knowledge	●			●		●	●	●	●
Two separate control rooms, no centralised control				●		●		●	●
Service continuity as first objective (goal conflict)		●	●				●	●	●
Missing interlocks / protections designed in isolation		●		●	●	●	●	●	●
Undersized discharge, pumping room not isolated			●		●				
Night shift, low attention, fatigue	●	●	●	●	●	●	●	●	●

Mapping based on this report; ● = factor appears in the template or in the justification of the event.

Key lessons

- **Dependent holes.** The same few latent factors (procedures, HMI, training, organisation of control rooms, service priority) reappear in almost every event: the barriers of the defence-in-depth were not independent, which is exactly what Reason's model predicts for organisational accidents.
- **Integrity of information.** The whole sequence starts from a false but plausible signal. A cyber attack on the integrity of sensor data could produce the same scenario: safety engineering and IT/OT security must stay aligned during the whole operating life, not only at design [Class].
- **Automation and situational awareness.** The automatic routine acted without involving the operators; when they were called into play they lacked the awareness needed for a correct diagnosis. Error prevention, error recovery and emergency management must be designed together.
- **Technical and organisational design.** Tank sizing, interlocks and alarms are effective only if aligned with procedures, training and response times (Event 3 discussion [Class]).
- **After the accident.** Re-design of the pumping stations, relocated and redundant sensors, new DCS, a single control room for generation and network, and a five-year programme on procedures and retraining [Class].